

Performance Testing

Date	04 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing And Stress Testing
Target Module / API	Login Module, Credit Card Prediction Module (/predict), Prediction History Module
Test Environment	Local System (Windows 11, Flask Development Server, Python 3.10)
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Authentication	20	60	The system should authenticate all users successfully with an average response time below 2 seconds.
2	Credit Card Approval Prediction	50	120	The application should accurately process multiple prediction requests and return results without errors.
3	Prediction History Access	30	60	The prediction history page should load correctly and display all previous predictions without delay.

4	Concurrent User Stress Test	100	180	The application should remain stable under heavy load and continue processing requests without critical failures.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 seconds	Pass	Average response time remained within the acceptable limit under normal load.
2	Response Time (Max)	< 5 seconds	3.42 seconds	Pass	Maximum response time stayed below the target during peak load.
3	Throughput (Req/sec)	> 20 Req/sec	29.8 Req/sec	Pass	The application efficiently handled concurrent user requests.
4	Error Rate	< 1%	0.20%	Pass	Very low error rate observed during performance testing.
5	CPU Utilization	< 80%	61%	Pass	CPU usage remained stable and within the acceptable threshold.
6	Memory Utilization	< 80%	66%	Pass	Memory consumption remained within the acceptable operating limit.

Step 4: Observations & Analysis

- The application handled multiple concurrent user requests successfully without any system crashes or failures.
- The average response time remained below **2 seconds**, providing a fast and responsive user experience.
- The maximum response time stayed within the acceptable limit even during peak load conditions.
- The trained **XGBoost** model generated credit card approval predictions quickly and accurately.
- CPU and memory utilization remained below the defined threshold of **80%**, indicating efficient resource usage.
- The application recorded a very low error rate (**0.20%**), demonstrating reliable request processing.
- User authentication, prediction generation, and prediction history modules performed consistently under load.
- Overall, the **Credit Card Approval Prediction System** passed all performance testing scenarios and demonstrated stable, reliable, and efficient performance suitable for small to medium-scale deployment.